

MAT2041 Assignment: Linear Transformation & Eigenvalues

Covering Lectures 20–23 (Excluding Determinant)

Spring 2026

Total: 100 points, 7 problems.

1 Linear Transformation over $\mathbb{R}^2$

(4 + 6 + 6 = 16 points)

Set $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

- (a) Consider the transformation that performs a vertical shear mapping $\mathbf{e}_1 \mapsto \mathbf{e}_1 + 2\mathbf{e}_2$ and leaves $\mathbf{e}_2$ unchanged. Determine the matrix A that represents this transformation.
- (b) Consider the transformation that rotates every vector counter-clockwise by 45° . Determine the matrix B that represents this transformation.
- (c) Consider the transformation that *first* performs the vertical shear in (a) and *then* rotates every vector counter-clockwise by 45° . Determine the matrix C that represents this transformation using A and B , and compute C explicitly.

2 Linear Transformation (Matrix Representation)

(12 points)

Let $\mathbb{P}_n$ be the vector space of all polynomials of degree at most n . Let

$$T : \mathbb{P}_2 \rightarrow \mathbb{P}_3$$

be the linear transformation defined by

$$(T(p))(x) = (x^2 - 2)p'(x), \quad \text{for all } p \in \mathbb{P}_2.$$

Find a matrix representation A for the linear transformation T .

3 Matrix Representation on Matrix Space

(5 + 10 = 15 points)

Define $T: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ by $T(X) = X + X^\top$.

- (a) Verify that T is a linear transformation.
- (b) Using the standard basis of $\mathbb{R}^{2 \times 2}$:

$$E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad E_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

find the 4×4 matrix representation A of T with respect to this basis (used for both the domain and the codomain).

4 Eigenvalue Decomposition & Diagonalisation

(8 + 8 = 16 points)

Given

$$A = \begin{bmatrix} 2 & -4 & -4 \\ -4 & -1 & -1 \\ -4 & -1 & -1 \end{bmatrix}.$$

(a) Compute the eigenvalues and corresponding eigenvectors of A .

(b) Find an orthogonal matrix P such that $P^\top AP = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$ with $\lambda_1 \geq \lambda_2 \geq \lambda_3$.

5 Eigenvalues of Inverse Matrix

(6 + 6 = 12 points)

Suppose $A \in \mathbb{R}^{n \times n}$ is invertible.

- (a) Prove that if λ is an eigenvalue of A , then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .
- (b) Prove that the multiset of eigenvalues of A^{-1} is $\#\{\lambda_1^{-1}, \dots, \lambda_n^{-1}\}$, where $\#\{\lambda_1, \dots, \lambda_n\}$ is the multiset of eigenvalues of A .
(hint: using characteristic polynomial)

6 Eigenvalues (Structural Properties)

(5 + 5 + 5 = 15 points)

1. Let A be an $n \times n$ matrix with an eigenvalue λ and corresponding eigenvector $\mathbf{x}$. Prove that for any scalar c , the matrix $(A + cI)$ has an eigenvalue $(\lambda + c)$ with the same eigenvector $\mathbf{x}$.
2. Let J be the 3×3 matrix where every entry is 1. Find the eigenvalues of J by computing its characteristic polynomial.
3. Let

$$C = \begin{bmatrix} 5 & 2 & 2 \\ 2 & 5 & 2 \\ 2 & 2 & 5 \end{bmatrix}.$$

Using your theoretical results from Part 1 and Part 2, find the eigenvalues of C **without** computing its characteristic polynomial.

7 Eigenvalues of Matrix Product

(7 + 7 = 14 points)

Let $A, B \in \mathbb{R}^{n \times n}$.

- (a) If $\lambda \neq 0$ is an eigenvalue of AB , prove that λ is also an eigenvalue of BA .
- (b) If $\lambda = 0$ is an eigenvalue of AB , prove that $\lambda = 0$ is also an eigenvalue of BA .